Modeling Wind Drift in Urban Drone Routes

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1 Introduction

1.1 Personal Connections and descriptions

The living experiences in cities are completely different because cities are three-dimensional, dynamic environments composed of buildings, streets, and lakes, and they face natural disasters such as floods and typhoons. Having lived in Shenzhen my entire life, I have witnessed its tremendous development over the past thirty years, as it changed from a small fishing village into a magnificent city that ranks at the top of my mind. Watching tall buildings, giant shopping centers, beautiful parks, and other public spaces being built from the bottom up, I felt deeply proud of how Shenzhen has developed. This city always embraces everyone with joy, provides energy to work, and brings hope to our daily lives.

However, my imagination of an ideal city changed from being solely magnificent to including diverse cultures and other factors after I came to Boston for high school. Boston is a city shaped by a calm atmosphere, historic street patterns, and academic energy. Having lived in both cities for a long time, I find that the geographic layouts of the two cities are completely different: Shenzhen is characterized by tall, crowded buildings, while Boston is characterized by a hybrid of low- and medium-sized buildings. These geographic factors significantly influence people in real life, including people's travel, vehicles' movement, and even how wind shifts or flows.

1.2 Study Topics

Facing buildings of various sizes, there are many possibilities and variations in wind: slowing or speeding up, moving forward or backward, and intensifying or weakening. Thus, a drone flying in the air does not simply travel from destination A to destination B. Instead, it requires a carefully planned route that accounts for wind variations, building heights, and real-time conditions to ensure safety and efficiency.

This project studies the city's layout and its influence on urban drone route planning, using differential equations and linear algebra. The comparison between two completely different cities, Boston, Massachusetts, and Shenzhen, China, provides insights for urban planning. To make the comparison even more specific, I focus on two representative and favorite drone routes: from Harvard to MIT in Boston and from Spring Bamboo to Poly Theater in Shenzhen. They are approximately the same length, around 1.4 miles, but the urban layouts are completely different. These two routes provide insights into how different urban forms may influence wind drift along actual city paths. The geographic differences suggest that a drone route might experience variation due to different wind disturbances in both cities. Therefore, the main research question of this project is:

> *How do different urban layouts affect wind drift and the flight paths of drones in cities?*

Buildings, city blocks, and hills are all potential factors that can influence the speed and direction of winds. Since drones can be used for delivery, traffic monitoring, and emergency situations, understanding wind drift is important for city planners who need to determine optimal and safe routes for drones.

1.3 Course Topics Used

This project uses mathematical ideas from Chapters 1–8 of *Differential Equations & Linear Algebra*.

- **Chapter 1: First-Order Differential Equations.** The drone motion model is a first-order differential equation because it describes how position changes with time. Slope fields and solution curves can be used to visualize how wind direction changes across a city.
- **Chapter 2: Mathematical Models and Numerical Methods.** Realistic city wind fields are too complicated to solve exactly, so Euler's method and the Runge-Kutta method are used to approximate the flight path step by step.
- **Chapter 3: Linear Systems and Matrices.** A city map can be represented by matrices: one matrix for building heights, one for building density, and one for risk intensity. Matrix operations help organize the data.
- **Chapter 4: Vector Spaces.** Wind and drone velocity are vectors in $\mathbb{R}^3$. A vector field assigns a wind vector to every location in the city.

The main mathematical foundation is Chapter 1 plus Chapter 2. The linear algebra chapters make the model more organized because the city is naturally stored as gridded data.

2 Mathematical Background

2.1 Linear vs. Non-linear

In this section, I will introduce some fundamental theorems and mathematical concepts useful for understanding the project. Since drones are easily influenced by wind shifts, I approach this project from both simplified and realistic perspectives. A linear system represents a simplified version, while a non-linear system depicts more complex and realistic situations. In a linear system, we assume the wind changes at a constant speed, whereas in a non-linear system, we assume its speed and direction vary with building height, street layout, and the surrounding urban environment. Comparing the ideal and realistic cases provides me with a comprehensive understanding of how urban layouts can influence wind drift and, in turn, change the drone's flight routes.

2.2 Mathematical Formulas

2.2.1 Linear System

Matrix model:

$$\mathbf{r}'(t) = A\mathbf{r}(t) + \mathbf{b} \tag{1}$$

Vector addition:

$$\mathbf{v}_{\text{actual}} = \mathbf{v}_{\text{drone}} + \mathbf{v}_{\text{wind}} \tag{2}$$

Urban wind-risk model:

$$U(x, y) = \alpha H(x, y) + \beta D(x, y) + \gamma C(x, y) \tag{3}$$

2.2.2 Non-Linear System

First-order nonlinear wind model:

$$\begin{aligned} \frac{dx}{dt} &= v_x + W_x(x, y) \\ \frac{dy}{dt} &= v_y + W_y(x, y) \end{aligned} \tag{4}$$

Euler's Method:

$$y_{n+1} = y_n + hf(t_n, y_n) \tag{5}$$

Improved Euler's Method:

$$y_{n+1} = y_n + \frac{h}{2}\left[f(t_n, y_n) + f\left(t_n + h,\ y_n + hf(t_n, y_n)\right)\right] \tag{6}$$

Runge-Kutta Method:

$$\begin{aligned}
k_1 &= f(t_n, y_n) \\
k_2 &= f\left(t_n + \frac{h}{2},\ y_n + \frac{h}{2}k_1\right) \\
k_3 &= f\left(t_n + \frac{h}{2},\ y_n + \frac{h}{2}k_2\right) \\
k_4 &= f(t_n + h,\ y_n + hk_3) \\
y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4)
\end{aligned} \tag{7}$$

3 Problem Setup

3.1 Variable Definitions

Symbol	**Meaning**
$\mathbf{r}(t)$	Actual drone position at time t.
$\mathbf{r}_0(t)$	Ideal no-wind straight flight path.
$x(t)$	Horizontal position of the drone at time t.
$y(t)$	Vertical position of the drone at time t.
t	Time variable.
t_n	Time value at the nth numerical step.
h	Step size used in Euler's method, improved Euler's method, and Runge-Kutta method.
n	Step number in the numerical approximation.
$\mathbf{v}_{\text{drone}}$	Drone's intended velocity without wind.
$\mathbf{v}_{\text{wind}}$	Wind velocity affecting the drone.
$\mathbf{v}_{\text{actual}}$	Actual drone velocity after combining drone velocity and wind velocity.
v_x	Drone's intended velocity in the x-direction.
v_y	Drone's intended velocity in the y-direction.

Symbol	Meaning
$\mathbf{W}(x,y)$	Local wind vector field at position (x,y).
$W_x(x,y)$	Wind effect in the x-direction at position (x,y).
$W_y(x,y)$	Wind effect in the y-direction at position (x,y).
A	Matrix representing the linear wind effect from the urban layout.
$\mathbf{b}$	Constant velocity vector representing the drone's intended motion.
$H(x,y)$	Normalized building height field at position (x,y).
$D(x,y)$	Normalized building density field at position (x,y).
$C(x,y)$	Normalized street-canyon intensity field at position (x,y).
$U(x,y)$	Combined urban wind-risk intensity at position (x,y).
α	Weight assigned to building height in the urban-risk model.
β	Weight assigned to building density in the urban-risk model.
γ	Weight assigned to street-canyon intensity in the urban-risk model.
$E(t)$	Distance between the actual route and the ideal no-wind route at time t.
η	Route efficiency score.
y_n	Approximate value of the dependent variable at step n.
y_{n+1}	Approximate value of the dependent variable at the next step.
$f(t_n, y_n)$	Slope function used in Euler's method and Runge-Kutta methods.
k_1, k_2, k_3, k_4	Intermediate slopes used in the fourth-order Runge-Kutta method.
P_0	Starting point of the drone route.
P_f	Final destination of the drone route.

3.2 Mathematical Application

To discuss whether urban layouts can influence wind drift and, in turn, the drone's flight route, I organize the situation in two ways. The first is represented by a linear system that accounts only for wind speed and drone speed. I used vector addition and a linear system model to achieve the idea, shown in (1) and (2). In addition, to compare whether Boston or Shenzhen exerts a stronger wind-drift influence on drones, I constructed a linear combination of matrices, where each matrix represents height, density, and street intensity

(3). After assigning weights to different matrices, the final combined results will indicate which city has the stronger influence on drones via wind.

However, urban effects, such as building heights, area density, and open spaces, will significantly influence wind speed and direction, which in turn affect the drone's flight route. As a result, we will use Euler's method(5), improved Euler's method(6), and the Runge-Kutta method (7) to estimate the flight paths of drones affected by urban effects.

3.3 Assumptions

Since wind speed and direction vary from time to time, they are unlikely to remain constant or perfectly predictable. In addition, cities around the world have different policies regarding drone flying heights. Therefore, throughout this paper, I make several reasonable assumptions to reduce the influence of confounding variables and focus primarily on wind shifts caused by urban layouts.

1. When creating graphs comparing ideal and realistic drone routes using Euler's Method, Improved Euler's Method, and the Runge-Kutta Method, I assume that the ideal route experiences no wind drift.
2. Since the maximum drone flying height varies across cities, I assume a flying height of 265 feet in both Boston and Shenzhen.
3. I assume the distance from Harvard University to MIT in Boston is approximately 2.6 kilometers, and the distance from Spring Bamboo to Poly Theatre in Shenzhen is approximately 1.5 kilometers.
4. I use a total of 20 steps to estimate the drone routes from Harvard University to MIT and from Spring Bamboo to Poly Theatre using Euler's Method, Improved Euler's Method, and the Runge-Kutta Method. Using 20 steps provides a reasonable balance between calculation simplicity and numerical accuracy.
5. I assume Harvard University as the origin

$$(0, 0)$$

and MIT as approximately

$$(2.05, -1.60)$$

This means that MIT is treated as 2.05 km east and 1.60 km south of Harvard University.

6. I assume Spring Bamboo as the origin

$$(0,0)$$

and Poly Theatre as approximately

$$(1.45, 0.35)$$

This means that Poly Theatre is treated as 1.45 km east and 0.35 km north of Spring Bamboo.

7. I use a wind speed of

$$3 \text{ m/s} \approx 6.6 \text{ mph}$$

and a drone speed of

$$10 \text{ m/s} \approx 22.3 \text{ mph}$$

in both the Boston and Shenzhen cases.

8. I assume α=0.5, β=0.3, and γ=0.2 in the weighted linear combination equation 3. These values hold true for both Boston and Shenzhen.

4 Mathematical Analysis

4.1 Linear System

In this section, I will use two methods to solve the linear system, including a combination of matrix model (1) and vector addition (2), and urban wind-risk model (3).

4.1.1 Urban wind-risk model

The urban wind-risk model (3) is a weighted linear combination equation used to measure how strongly the surrounding city layout may affect wind drift along a drone route. I specifically compare the city layout in Cambridge, where Harvard and MIT are located, to Nanshan, where Spring Bamboo and Poly Theatre are situated. Since buildings and streets can change wind speed and direction, the model combines three important urban factors: building height, building density, and street-canyon intensity. Each factor is normalized between 0 and 1, so they can be compared on the same scale.

$$H = \frac{\text{average building height near route}}{\text{maximum reference height}}$$

$$D = \frac{\text{building footprint area near route}}{\text{total route-buffer area}}$$

$$C = \frac{\text{average nearby building height}}{\text{average street width}}$$

The final value $U(x, y)$ represents the overall urban wind-risk score at location (x, y). Finally, we can compare the generated score between two cities and further understand which one is more influenced by urban effects. Since there are no direct data on building height, building density, and street-canyon intensity, I estimate these metrics from geographic information for Cambridge and Shenzhen. The estimated values are shown in Table 2.

Route	$H(x, y)$	$D(x, y)$	$C(x, y)$	$U(x, y)$
Harvard to MIT	0.25	0.45	0.35	0.330
Spring Bamboo to Poly Theatre	0.85	0.75	0.70	0.790

Table 2: Estimated normalized urban wind-risk values for the two routes.

For the calculation, I use the conditions I assumed in Section 3.3, which are the weighted values of α, β, and γ.

$$U_{\text{Boston}} = 0.5(0.25) + 0.3(0.45) + 0.2(0.35) = 0.330$$

$$U_{\text{Shenzhen}} = 0.5(0.85) + 0.3(0.75) + 0.2(0.70) = 0.790$$

4.1.2 Vector addition

The actual velocity of the drone is calculated by adding the drone's original velocity and the wind velocity:

$$\mathbf{v}_{actual} = \mathbf{v}_{drone} + \mathbf{v}_{wind}$$

I previously assumed wind speed in Section 3.3 as

$$3 \text{ m/s} = 0.18 \text{ km/min}$$

Since the urban wind-risk score changes the strength of the wind effect, I define

$$\mathbf{v}_{wind} = 0.18U(x, y)\hat{w}$$

where $\hat{w}$ represents the wind-direction unit vector.

We previously calculated that the Harvard-to-MIT route is

$$U_{\text{Boston}} = 0.330$$

Therefore, the speed of wind disturbance is Cambridge area is

$$|\mathbf{v}_{wind,B}| = 0.18(0.330) = 0.0594 \text{ km/min}$$

We previously calculated that the Spring Bamboo-to-Poly Theatre route is

$$U_{\text{Shenzhen}} = 0.790$$

Therefore, the speed of wind disturbance in Nanshan district is

$$|\mathbf{v}_{wind,S}| = 0.18(0.790) = 0.1422 \text{ km/min}$$

Therefore, vector addition shows that the wind-adjusted velocity is larger in Shenzhen than in Boston, which supports the idea that the Shenzhen route experiences stronger wind drift.

4.2 Nonlinear System

In a realistic perspective, the wind and urban effects are not constant. As a result, in this section, I will use three different methods to approximate the drone path and compare it with the ideal path. By using a total step of 20, I compare Euler(5), Improved Euler (6) and Runge-Kutta method (7) with the idea path.

4.2.1 Harvard → MIT

For the Harvard-to-MIT route, the wind field is modeled by

$$\frac{dx}{dt} = 0.4730 + 0.0080 \sin\left(\frac{\pi x}{2.05}\right)$$

$$\frac{dy}{dt} = -0.3692 + 0.0103 \sin\left(\frac{\pi x}{2.05}\right)$$

Table 3 summarizes the final numerical approximations for the Harvard-to-MIT route.

Method	Final Point	Drift from MIT
Euler's Method	$(2.0719, -1.5720)$	35.55 m
Improved Euler's Method	$(2.0718, -1.5720)$	35.50 m
Runge-Kutta Method	$(2.0719, -1.5720)$	35.57 m

Table 3: Final numerical approximations for the Harvard-to-MIT route.

A complete 20-step position dataset by using Euler, Improved Euler, and Runge-Kutta methods is shown in Table (4)

Table 4: Complete 20-step position dataset for the Harvard-to-MIT route.

Step	t	No Wind Route		Euler's Method		Improved Euler's Method		Runge-Kutta Method	
	min	x	y	x	y	x	y	x	y
0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.2167	0.1025	-0.0800	0.1025	-0.0800	0.1026	-0.0798	0.1026	-0.0798
2	0.4334	0.2050	-0.1600	0.2053	-0.1597	0.2055	-0.1593	0.2055	-0.1593
3	0.6501	0.3075	-0.2400	0.3083	-0.2390	0.3087	-0.2385	0.3087	-0.2384
4	0.8668	0.4100	-0.3200	0.4116	-0.3179	0.4121	-0.3173	0.4121	-0.3173
5	1.0835	0.5125	-0.4000	0.5151	-0.3966	0.5158	-0.3958	0.5158	-0.3958
6	1.3002	0.6150	-0.4800	0.6189	-0.4750	0.6196	-0.4741	0.6196	-0.4741
7	1.5169	0.7175	-0.5600	0.7228	-0.5532	0.7236	-0.5522	0.7236	-0.5522
8	1.7337	0.8200	-0.6400	0.8268	-0.6312	0.8277	-0.6302	0.8277	-0.6301
9	1.9504	0.9225	-0.7200	0.9310	-0.7091	0.9319	-0.7080	0.9319	-0.7080
10	2.1671	1.0250	-0.8000	1.0352	-0.7869	1.0361	-0.7858	1.0361	-0.7857
11	2.3838	1.1275	-0.8800	1.1395	-0.8647	1.1403	-0.8636	1.1404	-0.8635
12	2.6005	1.2300	-0.9600	1.2437	-0.9425	1.2445	-0.9414	1.2445	-0.9414
13	2.8172	1.3325	-1.0400	1.3478	-1.0204	1.3486	-1.0194	1.3486	-1.0193
14	3.0339	1.4350	-1.1200	1.4519	-1.0984	1.4526	-1.0975	1.4526	-1.0974
15	3.2506	1.5375	-1.2000	1.5558	-1.1766	1.5564	-1.1758	1.5564	-1.1758
16	3.4673	1.6400	-1.2800	1.6594	-1.2551	1.6599	-1.2545	1.6600	-1.2544
17	3.6840	1.7425	-1.3600	1.7629	-1.3338	1.7633	-1.3333	1.7633	-1.3333
18	3.9007	1.8450	-1.4400	1.8662	-1.4129	1.8664	-1.4126	1.8665	-1.4125
19	4.1174	1.9475	-1.5200	1.9692	-1.4923	1.9693	-1.4921	1.9693	-1.4921
20	4.3341	2.0500	-1.6000	2.0719	-1.5720	2.0718	-1.5720	2.0719	-1.5720

4.2.2 Spring Bamboo → Poly Theatre

For the Spring-Bamboo-to-Poly-Theatre route, the wind field is modeled by

$$\frac{dx}{dt} = 0.5832$$

$$\frac{dy}{dt} = 0.1408 + 0.0708 \sin\left(\frac{2\pi x}{1.45}\right)$$

Table 5 summarizes the final numerical approximations for the Spring-Bamboo-to-Poly-Theatre route.

Method	Final Point	Maximum Deviation from No-Wind Route
Euler's Method	$(1.4500, 0.3500)$	55.54 m
Improved Euler's Method	$(1.4500, 0.3500)$	55.54 m
Runge-Kutta Method	$(1.4500, 0.3500)$	56.00 m

Table 5: Final numerical approximations for the Spring-Bamboo-to-Poly-Theatre route.

A complete 20-step position dataset using Euler's Method, Improved Euler's Method, and the Runge-Kutta Method is shown in Table 6.

Table 6: Complete 20-step position dataset for the Spring-Bamboo-to-Poly-Theatre route.

Step	t	No Wind Route		Euler's Method		Improved Euler's Method		Runge-Kutta Method	
	min	x	y	x	y	x	y	x	y
0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.1243	0.0725	0.0175	0.0725	0.0175	0.0725	0.0189	0.0725	0.0189
2	0.2486	0.1450	0.0350	0.1450	0.0377	0.1450	0.0403	0.1450	0.0403
3	0.3729	0.2175	0.0525	0.2175	0.0604	0.2175	0.0639	0.2175	0.0640
4	0.4972	0.2900	0.0700	0.2900	0.0850	0.2900	0.0892	0.2900	0.0893
5	0.6215	0.3625	0.0875	0.3625	0.1109	0.3625	0.1153	0.3625	0.1155
6	0.7458	0.4350	0.1050	0.4350	0.1372	0.4350	0.1414	0.4350	0.1417
7	0.8701	0.5075	0.1225	0.5075	0.1630	0.5075	0.1666	0.5075	0.1670
8	0.9944	0.5800	0.1400	0.5800	0.1876	0.5800	0.1902	0.5800	0.1907
9	1.1187	0.6525	0.1575	0.6525	0.2103	0.6525	0.2117	0.6525	0.2121
10	1.2430	0.7250	0.1750	0.7250	0.2305	0.7250	0.2305	0.7250	0.2310
11	1.3673	0.7975	0.1925	0.7975	0.2480	0.7975	0.2467	0.7975	0.2471
12	1.4916	0.8700	0.2100	0.8700	0.2628	0.8700	0.2602	0.8700	0.2607
13	1.6159	0.9425	0.2275	0.9425	0.2751	0.9425	0.2716	0.9425	0.2720
14	1.7403	1.0150	0.2450	1.0150	0.2855	1.0150	0.2814	1.0150	0.2817
15	1.8646	1.0875	0.2625	1.0875	0.2947	1.0875	0.2903	1.0875	0.2905
16	1.9889	1.1600	0.2800	1.1600	0.3034	1.1600	0.2992	1.1600	0.2993
17	2.1132	1.2325	0.2975	1.2325	0.3125	1.2325	0.3089	1.2325	0.3090
18	2.2375	1.3050	0.3150	1.3050	0.3229	1.3050	0.3203	1.3050	0.3203
19	2.3618	1.3775	0.3325	1.3775	0.3352	1.3775	0.3339	1.3775	0.3339
20	2.4861	1.4500	0.3500	1.4500	0.3500	1.4500	0.3500	1.4500	0.3500

5 Interpretation and Discussion

5.1 Analysis of results

The mathematical results show that urban layout has a clear influence on drone wind drift. Using the urban wind-risk model in (3), the Harvard-to-MIT route has a risk score of

$$U_{\text{Boston}} = 0.330$$

while the Spring-Bamboo-to-Poly-Theatre route has a much higher risk score of

$$U_{\text{Shenzhen}} = 0.790$$

This means that, under the assumptions of this project Section 3.3, the Shenzhen route experiences a stronger urban wind effect than the Boston route. The main reason is that Shenzhen has taller buildings, higher density, and stronger street-canyon effects, while Boston has a lower and more compact urban structure.

The vector addition results also support this conclusion. Since the wind disturbance is modeled as

$$|\mathbf{v}_{\text{wind}}| = 0.18U(x, y)$$

the Boston wind disturbance is only

$$0.18(0.330) = 0.0594 \text{ km/min}$$

while the Shenzhen wind disturbance is

$$0.18(0.790) = 0.1422 \text{ km/min}$$

If we compare these two numbers, the wind disturbance score in the Shenzhen route is twice as large as the disturbance along the Boston route. This suggests that drones flying in denser districts may require careful path planning consideration.

5.2 Ideas behind approximation methods

The numerical methods show how wind drift changes the actual flight path. For the Harvard-to-MIT route, using Euler, Improved Euler, and Runge-Kutta method all reach destination points closer to MIT, with a drift of about 35 to 36 meters. This indicates that Boston's urban layout still affects the drone, but the effect is relatively small. For the Spring-Bamboo-to-Poly-Theatre route, even though the final destination is reached, the deviation from the no-wind route is about 55 to 56 meters. This indicates that even if the drone ends up at the final destination, the flying path can still be pushed away from the ideal straight route. Even though the final destinations of the three approximation methods did not seem significantly different in this project, there are still significant differences among them. As a result, understanding the mathematical reasons behind Euler, Improved Euler, and Runge-Kutta is important. Euler's method provides a rough, step-by-step approximation because it only uses the slope at the start of each step. Improved Euler's Method uses the average of the two slopes, as shown in (6), yielding more accurate results than the Euler method. The Runge-Kutta method provides more information about the slope; averaging four slopes in a single step yields an even more reliable result than the previous methods. However, all three methods give similar overall conclusions: in both cities, urban wind effects push the drone away from the ideal straight-line path, and the effect is stronger in Shenzhen than in Boston.

5.3 Conclusion

Overall, the results suggest that drone route planning should not only consider distance. Even when two routes have similar lengths, the surrounding city structure can create very different wind-drift effects. A route through a dense high-rise area may require more advanced correction than a route through a lower-density area. This supports the main idea of the project: different urban layouts can meaningfully affect drone flight paths, especially in cities with taller buildings and stronger street-canyon conditions.

6 Graphical Analysis

6.1 City Map

I gathered data on Boston City-managed streets, building inventory, and Boston zoning subdistricts, then entered it and created the 3D graph. The three-dimensional graph shows the layout of Boston, dividing the buildings into 5 groups. Different-colored bars demonstrate the building heights. Below is the Boston City Map.

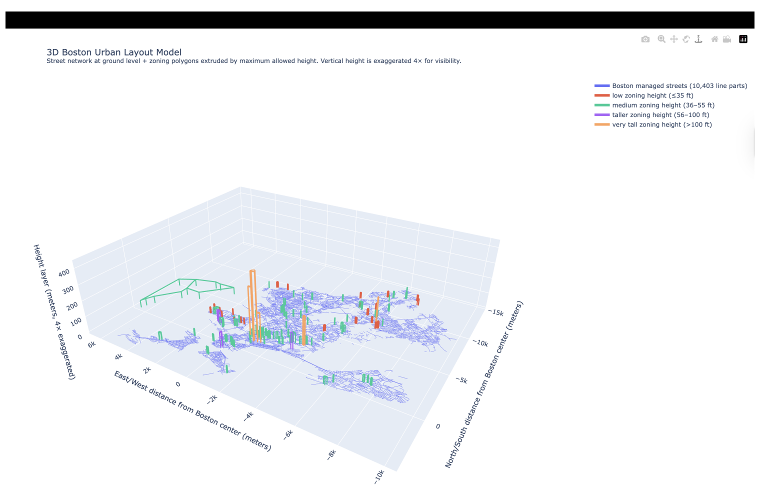

Figure 1: Boston City Layout

I gathered data on Shenzhen City's street network, building locations, and building heights, then entered it and created the 3D graph. The three-dimensional graph shows the layout of Boston, dividing the buildings into 5 groups. Different-colored bars demonstrate the building heights. Below is the Shenzhen city map.

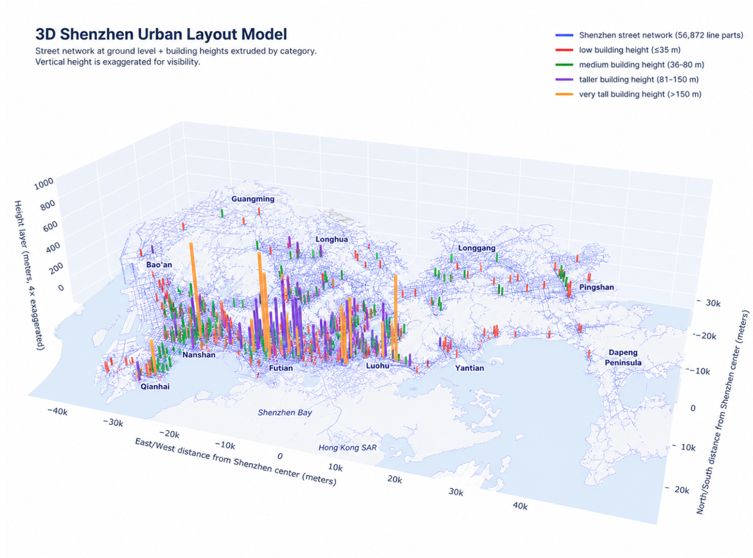

Figure 2: Shenzhen City Layout

Based on the two 3D urban layout models, Shenzhen shows a much larger and more vertically intense urban structure than Boston. Boston's map is relatively compact, with most height layers staying low or medium, and only a few taller zones appearing around the city center. Its street network looks dense but spread across a smaller area, suggesting a more moderate urban form with fewer extreme height changes. In contrast, Shenzhen has a much wider spatial scale and many more high-rise clusters, especially around Nanshan, Futian, and Luohu. The taller orange and purple extrusions indicate that Shenzhen's skyline is more vertically concentrated, with many buildings reaching above 80 meters and some exceeding 150 meters. This difference suggests that drone routes in Shenzhen may experience stronger urban wind-canyon effects, sudden wind shifts, and greater obstruction from tall buildings, while Boston's drone routes may be influenced more by street-grid density, lower zoning variation, and smaller-scale building clusters. Overall, Boston appears to be a lower, more compact urban environment, while Shenzhen appears to be a larger, denser, and more vertically complex city, making Shenzhen potentially more challenging for drone route planning.

6.2 Route Creation

6.2.1 Harvard → MIT

I used Euler, Improved Euler, and Runge-Kutta methods to approximate the flying route from Harvard to MIT. The dotted black line represents the straight path with no wind shift in the air. The red/blue line represents the targeted method estimated to be influenced by urban effects, including building heights and densities in the region.

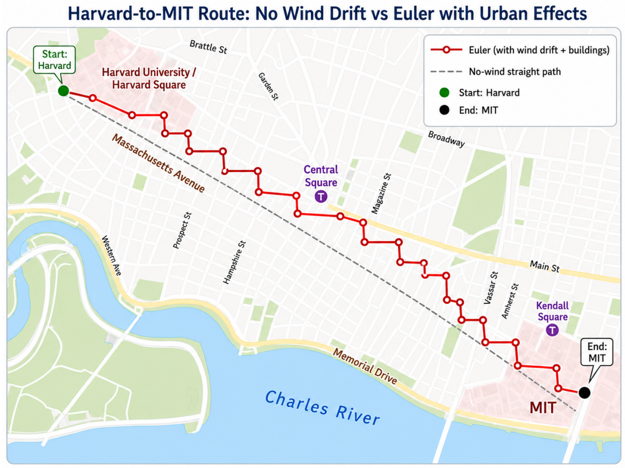

Figure 3: Route estimation from Harvard to MIT using Euler's method

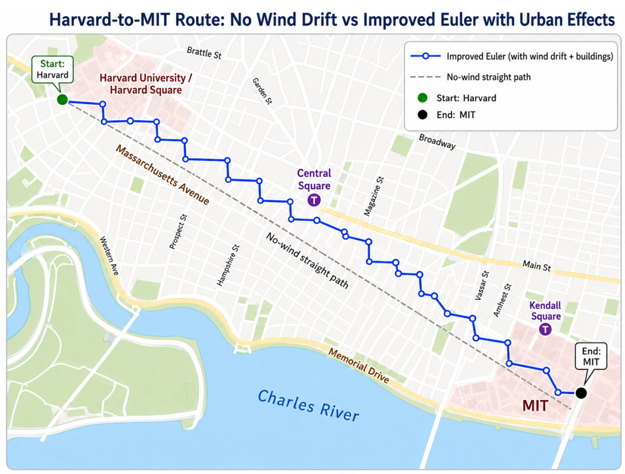

Figure 4: Route estimation from Harvard to MIT using Improved Euler's method

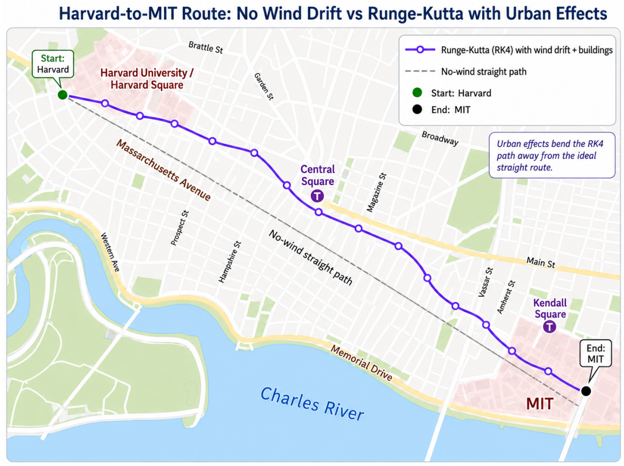

Figure 5: Route estimation from Harvard to MIT using the Runge-Kutta method

6.2.2 Spring Bamboo → Poly Theatre

I used Euler, Improved Euler, and Runge-Kutta methods to approximate the flying route from Spring Bamboo to Poly Theatre. The dotted black line represents the straight path with no wind shift in the air. The red/blue line represents the targeted method estimated to be influenced by urban effects, including building heights and densities in the region.

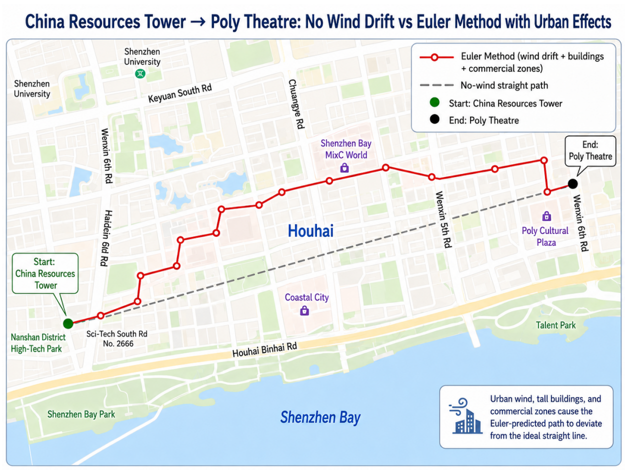

Figure 6: Route estimation from Spring Bamboo to Poly Theatre using Euler's method

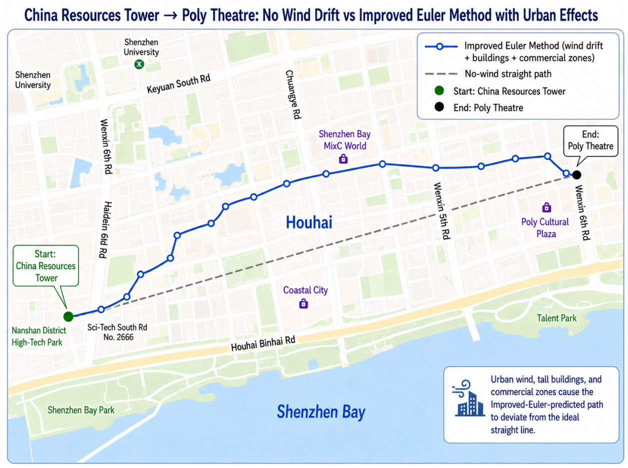

Figure 7: Route estimation from Spring Bamboo to Poly Theatre using Improved Euler's method

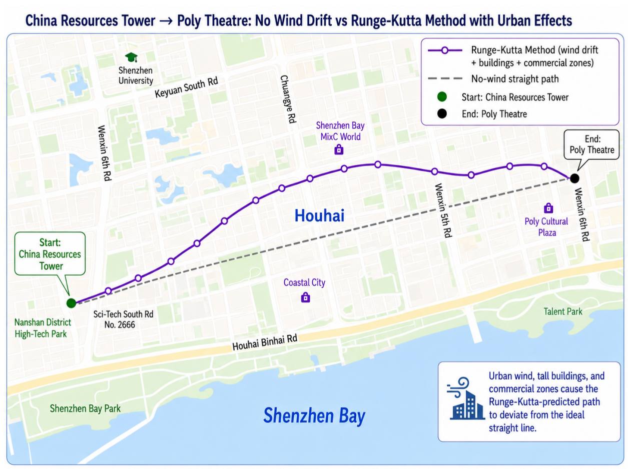

Figure 8: Route estimation from Spring Bamboo to Poly Theatre using the Runge-Kutta method

I created graphs comparing no wind drift with urban effects using Euler, Improved Euler, and Runge-Kutta methods. A no-wind straight path means drones can fly in a straight line, with no interference at all. However, when dealing with urban effects, I take into consideration tall buildings, such as MIT Kendall Square in Boston and Spring Bamboo in Shenzhen. These tall buildings will act as either a buffer or an active force on the wind, moderately altering wind speed and direction. These outputs show that wind and urban effects can push a drone away from the ideal straight path. Overall, Runge-Kutta gives the smoothest estimate, while Euler gives a rougher approximation.

6.3 Comparison of ideal straight-line drone route with straight-line with wind drift

6.3.1 Harvard → MIT

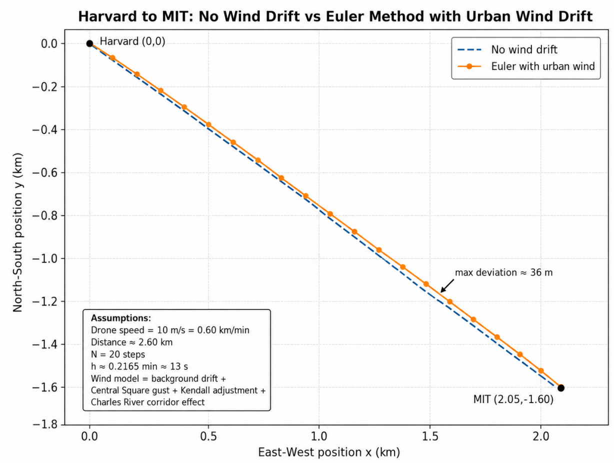

Figure 9: Harvard to MIT drone route with and without urban wind drift using Euler's Method

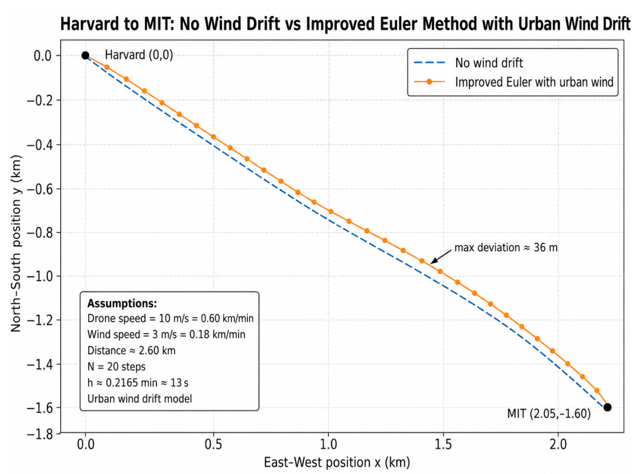

Figure 10: Harvard to MIT drone route with and without urban wind drift using Improved Euler's Method

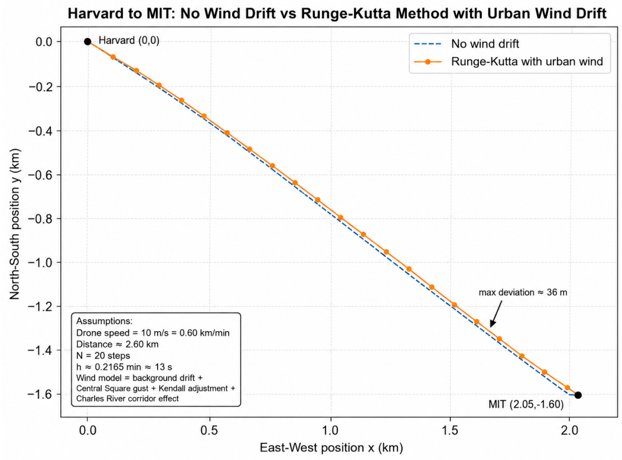

Figure 11: Harvard to MIT drone route with and without urban wind drift using Runge-Kutta Method

6.3.2 Spring Bamboo → Poly Theatre

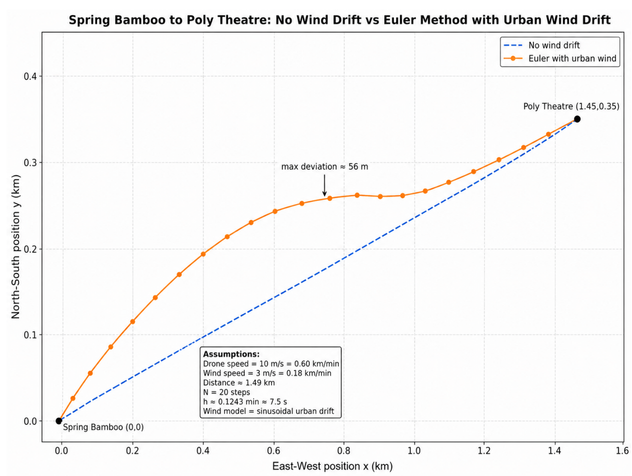

Figure 12: Spring Bamboo to Poly Theatre drone route with and without urban wind drift using Euler's Method

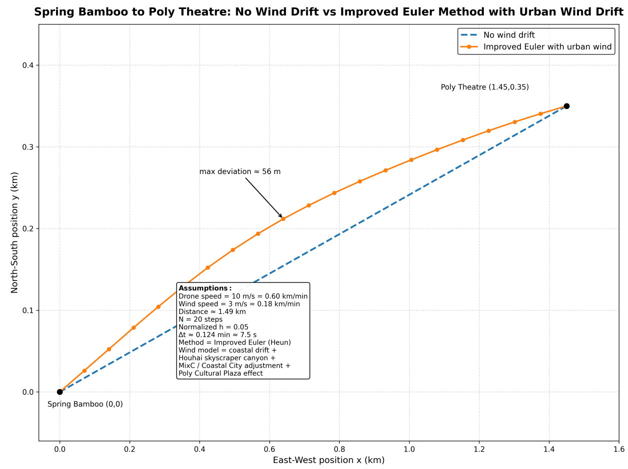

Figure 13: Spring Bamboo to Poly Theatre drone route with and without urban wind drift using Improved Euler's Method

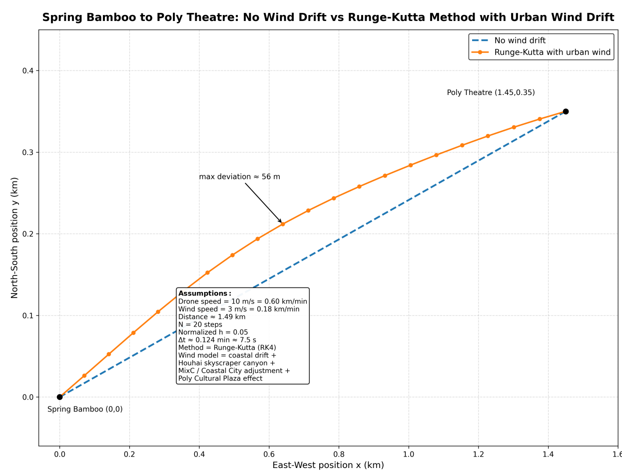

Figure 14: Spring Bamboo to Poly Theatre drone route with and without urban wind drift using Runge-Kutta Method

In the graph, using the conditions I assumed in Section 3.3, these graphs compare the ideal no-wind route with routes affected by urban wind drift in a plane. In both cities, the wind-drift paths move away from the straight baseline, showing that buildings and urban density can change the drone's flight direction. From Harvard to MIT, the deviation is relatively small, around 36 m. For Spring Bamboo to Poly Theatre, located in Nanshan region 2, the deviation is larger and more visible, around 56 m. This suggests that Shenzhen's denser and taller urban environment creates stronger drift effects than Boston.

7 Reflection

7.1 Key takeaways

Throughout this interesting project, most of the content was created by me. I spent at least 20 hours above and found so many intriguing ideas. This project provides me with opportunities to connect first-order differential equations, vector addition, the Euler, the Improved Euler, and the Runge-Kutta methods with urban studies. I learned that tall buildings, crowded streets, and high density will significantly influence the wind shift. This project also makes me feel connected to society by analyzing the Cambridge area and the Nanshan district, helping me better understand the areas around me. The drone's study indicates that different urban layouts will influence the city's future planning. More importantly, having been previously exposed to LaTeX , I operate smoothly and learned more commands this time. I really appreciate my teacher for offering the opportunity to learn and use LaTeX for the project, which has made me well prepared for college, where I will be required to submit neat, organized assignments.

7.2 Challenges

Throughout the project, I faced many challenges. I visited many government websites in both Boston and Shenzhen to review published data on building heights, regional density, and special regulations about the flying height and speed of drones in the city. Some of the datasets are not accessible, and some do not provide the information I want. In addition, most of the assumptions I made in Section 3.3 are limitations, including wind speed in both cities. Wind speed is constantly changing in both cities, so it is extremely difficult to find a common point of comparison. There are so many confounding variables that may contribute to the variations in the results section. If we want to specifically understand how urban effects, such as building height, may influence wind drift and thus affect the drone's flight path, we have to eliminate as many confounding variables as we can, which is nearly impossible in this easily designed project.

7.3 Potential improvements

There are some parts I can improve on if I want to dive deeper into the project. The first is to use real hourly wind data from weather stations near Cambridge and Shenzhen. This would provide more realistic and reliable data, since we previously assumed a wind speed of 3 m/s. Secondly, we can include more routes in both Boston and Shenzhen to see whether the result holds true for the whole city or only for this single selected path. Finally, when we are estimating the distance by using Euler, Improved Euler, and Runge-Kutta methods, we can include more steps for approximation, maybe 100 or more, to get more accurate results.

8 AI Usage Journal

8.1 Credits for AI

Since I really trust my best friend ChatGPT, I only used this AI tool throughout this project. I used ChatGPT 4.5 Thinking to partially assist me with LaTeX formatting. It suggested how to frame the project as an urban planning problem. ChatGPT also provided me with ideas to connect the topic to differential equations and linear algebra as tools for modeling wind drift and drone movement. In addition, ChatGPT 4.5 Thinking generated ideas and wrote some of the chapter descriptions for Section 1.3, the entire Section 3.1, some of the comparisons in Section 6.1, Section 6.3, and some of the discussion in Section 5.1 Section 5.3. It also helps me with the MATLAB coding for Euler, Improved Euler, and Runge-Kutta methods presented in Table (4) and Table 6.

8.2 Accuracy and usefulness of AI

For most of the time, AI calculations are correct. However, when they are calculating the linear combination using formula (3), they keep returning the wind-risk values for Harvard to MIT 0.335, but in reality it should be 0.330, shown in Table 2. For AI-generated content, I found most of it interesting and captured the conclusion I perceived. I'm sometimes just too lazy to analyze an obvious graph or write down conclusions by comparing some easy numbers, so I ask ChatGPT to write for me. I read all the content it wrote, for example Section 6.1, and revised or deleted some sentences that I felt were repetitive or other questions.

9 Reference page

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10 Code Appendix

10.1 MATLAB Code for Harvard to MIT Drone Route Approximation

```matlab
r0=[0;0];
rf=[2.05;-1.60];
N = 20;
droneSpeed=0.60;
windSpeed=0.18;
route=rf - r0;
distance=norm(route);
direction=route/distance;
T=distance/droneSpeed;
h=T/N;
vDrone=droneSpeed*direction;

fprintf('Distance: %.4f km\n', distance)
fprintf('Time: %.4f min\n', T)
fprintf('Step size: %.4f min\n\n', h)

time=linspace(0,T,N+1);

ideal=zeros(2,N+1);
euler=zeros(2,N+1);
improved=zeros(2,N+1);
rk4=zeros(2,N+1);

euler(:,1)=r0;
improved(:,1)=r0;
rk4(:,1)=r0;

for i=0:N
```

```
    ideal(:,i+1)=r0+(i/N)*(rf-r0);
end
for i=1:N
    wind=urbanWind(euler(:,i),windSpeed);
    euler(:,i+1)=euler(:,i)+h*(vDrone + wind);
end

for i=1:N
    k1=vDrone+urbanWind(improved(:,i),windSpeed);
    guess=improved(:,i)+h*k1;
    k2=vDrone+urbanWind(guess,windSpeed);
    improved(:,i+1)=improved(:,i)+(h/2)*(k1+k2);
end

for i = 1:N
    k1=vDrone+urbanWind(rk4(:,i),windSpeed);
    k2=vDrone+urbanWind(rk4(:,i)+(h/2)*k1,windSpeed);
    k3=vDrone+urbanWind(rk4(:,i)+(h/2)*k2,windSpeed);
    k4=vDrone+urbanWind(rk4(:,i)+h*k3,windSpeed);
    rk4(:,i+1)=rk4(:,i)+(h/6)*(k1+2*k2+2*k3+k4);
end

eulerDrift=norm(euler(:,end)-rf)*1000;
improvedDrift=norm(improved(:,end)-rf)*1000;
rk4Drift=norm(rk4(:,end)-rf)*1000;

fprintf('Final points:\n')
fprintf('Euler: %.6f, %.6f\n', euler(1,end), euler(2,end))
fprintf('Improved Euler: %.6f, %.6f\n', improved(1,end), improved(2,
    end))
fprintf('Runge-Kutta: %.6f, %.6f\n\n', rk4(1,end), rk4(2,end))

fprintf('Drift from MIT:\n')
fprintf('Euler: %.3f meters\n', eulerDrift)
fprintf('Improved Euler: %.3f meters\n', improvedDrift)
fprintf('Runge-Kutta: %.3f meters\n', rk4Drift)

Step=(0:N)';
Time=time';

```

```
ResultTable = table(Step,Time, ...
    ideal(1,:)',ideal(2,:)', ...
    euler(1,:)',euler(2,:)', ...
    improved(1,:)',improved(2,:)', ...
    rk4(1,:)',rk4(2,:)', ...
    'VariableNames',{'Step','Time','Ideal_x','Ideal_y', ...
    'Euler_x','Euler_y','Improved_x','Improved_y','RK4_x','RK4_y'});

disp(ResultTable)
figure
hold on
grid on

plot(ideal(1,:),ideal(2,:),'b--','LineWidth',2)
plot(euler(1,:),euler(2,:),'r-o','LineWidth',1.5)
plot(improved(1,:),improved(2,:),'m-s','LineWidth',1.5)
plot(rk4(1,:),rk4(2,:),'g-^','LineWidth',1.5)
plot(r0(1),r0(2),'ko','MarkerFaceColor','k')
plot(rf(1),rf(2),'ko','MarkerFaceColor','k')

text(r0(1)+0.04,r0(2),'Harvard')
text(rf(1)-0.25,rf(2)-0.05,'MIT')
xlabel('x position (km)')
ylabel('y position (km)')
title('Harvard to MIT Drone Route')
legend('No wind','Euler','Improved Euler','Runge-Kutta','Location','best')
axis equal

figure
hold on
grid on
plot(rf(1),rf(2),'ko','MarkerFaceColor','k')
plot(euler(1,end),euler(2,end),'ro','MarkerFaceColor','r')
plot(improved(1,end),improved(2,end),'ms','MarkerFaceColor','m')
plot(rk4(1,end),rk4(2,end),'g^','MarkerFaceColor','g')

xlabel('x position (km)')
ylabel('y position (km)')
title('Final Points Compared with MIT')
```

```
legend('MIT','Euler','Improved Euler','Runge-Kutta','Location','best
    ')
axis equal

function w = urbanWind(r,windSpeed)
x = r(1);
y = r(2);
wx = 0.10 + 0.16*sin(2*pi*x/2.05) ...
    + 0.10*exp(-((x-1.10)^2+(y+0.75)^2)/0.08);
wy = 0.12 + 0.20*sin(pi*x/2.05) ...
    + 0.08*cos(2*pi*y/1.60) ...
    + 0.10*exp(-((x-1.45)^2+(y+1.05)^2)/0.06);
w = windSpeed*[wx;wy];
if norm(w) > windSpeed
    w = windSpeed*w/norm(w);
end
end
```

10.2 MATLAB Code for Spring Bamboo to Poly Theatre Drone Route Approximation

```
r0=[0;0];
rf=[1.45; 0.35];
N=20;
droneSpeed=0.60;
windSpeed=0.18;
route=rf-r0;
distance=norm(route);
direction=route/distance;
T = distance/droneSpeed;
h=T/N;

vDrone = droneSpeed*direction;
fprintf('Distance=%.4f km\n', distance);
fprintf('Time=%.4f min\n', T);
fprintf('Step size=%.4f min\n\n', h);
% no-wind path
ideal=zeros(2, N+1);
for i=0:N
```

```
    ideal(:,i+1)=r0+(i/N)*route;
end
% set up arrays
euler=zeros(2, N+1);
improved=zeros(2, N+1);
rk4=zeros(2,N+1);
euler(:,1)=r0;
improved(:,1)=r0;
rk4(:,1)=r0;
time=linspace(0,T,N+1);
% Euler method
for i=1:N
    t=time(i);
    wind=shenzhenWind(t, euler(:,i), windSpeed, T);
    euler(:,i+1)=euler(:,i)+h*(vDrone+wind);
end
% Improved Euler method
for i=1:N
    t=time(i);
    k1=vDrone+shenzhenWind(t, improved(:,i), windSpeed, T);
    predict=improved(:,i)+h*k1;
    k2=vDrone+shenzhenWind(t+h, predict, windSpeed, T);
    improved(:,i+1)=improved(:,i)+(h/2)*(k1 + k2);
end
% Runge-Kutta 4 method
for i=1:N
    t=time(i);
    k1=vDrone+shenzhenWind(t, rk4(:,i), windSpeed, T);
    k2=vDrone+shenzhenWind(t+h/2, rk4(:,i)+(h/2)*k1, windSpeed, T);
    k3=vDrone+shenzhenWind(t+h/2, rk4(:,i)+(h/2)*k2, windSpeed, T);
    k4=vDrone+shenzhenWind(t+h, rk4(:,i)+h*k3, windSpeed, T);
    rk4(:,i+1) = rk4(:,i)+(h/6)*(k1+2*k2+2*k3+k4);
end

% final drift from Poly Theater
eulerDrift=norm(euler(:,end) - rf) * 1000;
improvedDrift=norm(improved(:,end) - rf) * 1000;
rk4Drift=norm(rk4(:,end) - rf) * 1000;

fprintf('Final points:\n');
```

```
fprintf('Euler:           (%.6f, %.6f)\n', euler(1,end), euler(2,end));
fprintf('Improved Euler: (%.6f, %.6f)\n', improved(1,end), improved(2,end));
fprintf('Runge-Kutta:    (%.6f, %.6f)\n\n', rk4(1,end), rk4(2,end));

fprintf('Drift from Poly Theater:\n');
fprintf('Euler drift = %.3f meters\n', eulerDrift);
fprintf('Improved Euler drift = %.3f meters\n', improvedDrift);
fprintf('Runge-Kutta drift = %.3f meters\n', rk4Drift);

% table
Step=(0:N)';
Time_min=time';

ResultTable=table(Step, Time_min, ...
    ideal(1,:)', ideal(2,:)', ...
    euler(1,:)', euler(2,:)', ...
    improved(1,:)', improved(2,:)', ...
    rk4(1,:)', rk4(2,:)', ...
    'VariableNames', {'Step','Time_min', ...
    'Ideal_x','Ideal_y', ...
    'Euler_x','Euler_y', ...
    'Improved_x','Improved_y', ...
    'RK4_x','RK4_y'});

disp(ResultTable);

% plot route
figure;
hold on;
grid on;

plot(ideal(1,:), ideal(2,:), 'b--', 'LineWidth', 2);
plot(euler(1,:), euler(2,:), 'r-o', 'LineWidth', 1.5);
plot(improved(1,:), improved(2,:), 'm-s', 'LineWidth', 1.5);
plot(rk4(1,:), rk4(2,:), 'g-^', 'LineWidth', 1.5);
plot(r0(1), r0(2), 'ko', 'MarkerFaceColor', 'k');
plot(rf(1), rf(2), 'ko', 'MarkerFaceColor', 'k');

```

```
text(r0(1)+0.04, r0(2), 'Spring Bamboo');
text(rf(1)-0.25, rf(2)+0.04, 'Poly Theater');
xlabel('x position (km)');
ylabel('y position (km)');
title('Spring Bamboo to Poly Theater Drone Route');
legend('No wind', 'Euler', 'Improved Euler', 'Runge-Kutta', ...
       'Location', 'best');
axis equal;
% plot final points
figure;
hold on;
grid on;

plot(rf(1), rf(2), 'ko', 'MarkerFaceColor', 'k');
plot(euler(1,end), euler(2,end), 'ro', 'MarkerFaceColor', 'r');
plot(improved(1,end), improved(2,end), 'ms', 'MarkerFaceColor', 'm')
   ;
plot(rk4(1,end), rk4(2,end), 'g^', 'MarkerFaceColor', 'g');
xlabel('x position (km)');
ylabel('y position (km)');
title('Final Points Compared with Poly Theater');
legend('Poly Theater', 'Euler', 'Improved Euler', 'Runge-Kutta', ...
       'Location', 'best');
axis equal;
% Shenzhen wind model
function w=shenzhenWind(t, r, windSpeed, T)
    x=r(1);
    y=r(2);
    wx=0.08 ...
        + 0.10 * sin(2*pi*t/T) ...
        + 0.08 * exp(-((x - 0.70)^2 + (y - 0.18)^2) / 0.05);
    wy=0.12 ...
        + 0.28 * sin(2*pi*t/T) ...
        + 0.12 * exp(-((x - 1.00)^2 + (y - 0.25)^2) / 0.04);
    w=windSpeed * [wx; wy];
    if norm(w) > windSpeed
        w=windSpeed * w / norm(w);
    end
end
```